



Balancing Privacy and Security of QNAME Minimisation

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Abstract

In the original design of DNS resolution, upstream authoritative servers have the potential to collect query privacy information, potentially resulting in user privacy leakage. To mitigate this, QNAME Minimisation (QMIN) is designed to provide enhanced privacy-preserving properties to the DNS resolver. QMIN exposes only the minimum sub-query needed for the authoritative server to guarantee that upstream nameservers only receive information about the sub-zone they directly manage. Nevertheless, QMIN may significantly increase the number of iterative queries and be exploited by adversaries to launch amplification attacks. Existing QMIN security mitigation strategies tend to reintroduce privacy vulnerabilities, highlighting the need for an enhanced solution that balances security mitigation and privacy protection.

In this work, we propose a quantitative model to assess both the privacy leakage and amplification potential across various QMIN strategies. We then carry out large-scale measurements to reveal the real-world deployment of QMIN strategies and their privacy and security risks, especially for those domains that have long labels or out-of-bailiwick referrals. Finally, we introduce Delegation Aware QMIN (DAQ), a novel QMIN strategy that effectively mitigates the

amplification vector while achieving a privacy enhancement similar to complete QMIN.

CCS Concepts

• **Security and privacy** → **Network security**.

Keywords

QNAME Minimisation; DNS Privacy; Denial-of-Service (DoS)

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Resource Availability:

Our data are publicly available at https://github.com/LOVELY-qinqin/www2026_QMIN_privacy_security (doi.org/10.5281/zenodo.18324210).

1 Introduction

The Domain Name System (DNS) provides domain name resolution services by mapping domain names to IP addresses [31][19][34]. In the original design of DNS, resolvers forward the entire queried domain name (QNAME) to upstream DNS nameservers during the resolution process [28]. However, transmitting the full QNAME has raised significant privacy concerns [20], as it allows upstream DNS nameservers to collect large amounts of sensitive user information during the resolution process [9]. For example, query logs maintained by Top-Level Domain (TLD) servers may reveal that users are requesting records for all subdomains of a given domain, thereby exposing detailed browsing patterns and behaviors [35][15].

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To minimise data exposure to mitigate privacy risks [14], QNAME Minimisation (QMIN) was introduced [10]. A DNS resolver that implements QMIN no longer sends the full QNAME to upstream DNS nameservers. Instead, it sends a QNAME containing only one additional label beyond the zone for which the server is authoritative, targeting the closest known ancestor of the original QNAME.

Although QMIN effectively protects the privacy of the client, it introduces additional amplification threat [17]. Attackers can exploit QMIN by constructing deeply-labeled QNAMEs, compelling resolvers to perform excessive iterative queries. RFC 9156 attempts to mitigate this by introducing the maximum iteration count per query [11]. However, DNS resolvers passively introduced additional QNAMEs during resolution. For example, when receiving responses with NS records without glue records, resolvers need to resolve the NS (Name Server) before resolving the original QNAME. And resolvers may apply QMIN not only to original QNAMEs but also to those introduced passively. Therefore, even with limits imposed, the passive introduction of numerous new QNAMEs requiring QMIN resolution can still overwhelm a resolver with a massive volume of queries.

To mitigate amplification potential, several mitigation strategies have been proposed, such as restricting QMIN to only the original QNAME or root & TLD servers. Such strategies reduce the attack surface, but reintroduce client privacy leakage [30]. Because referral QNAMEs often contain sensitive client information, and long QNAMEs can leak data to subzone nameservers (such as Second-Level Domains, or SLDs) during the original QNAME resolution process. So we need a quantitative model to evaluate privacy protection ability and amplification potential across various QMIN strategies. Based on this evaluation model and the results, our goal is to strike a better balance between the security and privacy provided by QMIN.

In this paper, we first develop a quantitative model to assess the security and privacy leakage of different QMIN strategies. Our analysis reveals that existing strategies perform poorly in balancing these two aspects: strategies offering strong privacy protection are highly vulnerable to amplification attacks, while those that minimise attack risk sacrifice substantial client privacy. By thoroughly investigating the source of privacy leakage, we introduce a new QMIN strategy called delegation-aware QMIN (DAQ). DAQ predicts the delegation point of each nameserver to reduce the iterative times per query and protect the client's privacy as much as possible. DAQ effectively reduces the attack potential close to the best existing anti-attack models (no QMIN) and maintains a privacy protection capability nearly identical to complete QMIN.

Contributions. In summary, our main contributions are as follows:

- *Modeling.* We propose a quantitative model to evaluate the privacy leakage and amplification ability of different QMIN strategies. This model consists of two parts: a privacy leakage quantifying model, which calculates the expected leakage amount per query, and an amplifying potential evaluation model, which assesses the security risk of the resolver.
- *Measurements.* We find that the amplification attack due to QMIN has a widespread impact on DNS servers. We perform measurements of QMIN deployment and find that more than 50.27% of the open resolvers and 21 of 39 public resolvers are

vulnerable to amplification attack. Furthermore, we find that 18,024 referral records may leak client privacy information to unrelated nameservers due to out-of-bailiwick delegation.

- *Optimization.* We propose a new strategy **DAQ** to better balance the privacy and security of QMIN. This approach effectively improves security and reduces the privacy leakage of the QNAME minimisation protocol, achieving an expected leakage amount nearly identical to complete QMIN.

2 Background and Motivation

2.1 Authoritative DNS Delegation & QMIN

DNS delegation underpins the DNS hierarchical and distributed architecture [32]. The DNS resolution process constitutes a chain of referrals guided by delegation points [25, 26]. In this architecture, upstream nameservers are only responsible for resolving referrals related to their managed sub-zones, not providing the final answer [7]. Nevertheless, sending a Fully Qualified Domain Name [22, 29], upstream nameservers could collect extra client information resolved by its sub-zone, causing unnecessary privacy leakage. QNAME minimisation (QMIN) is proposed to provide enhanced privacy-preserving properties to the resolver. QMIN exposes only the minimum sub-query needed to the authoritative server as it traverses the DNS delegation hierarchy to the target zone [19]. As of 2025, four major DNS service software solutions BIND9 [1], Unbound [4], Knot DNS [3], and PowerDNS [2] support QNAME. This guarantees that upstream nameservers only receive information about the sub-zone they directly manage, rather than the full original QNAME. For instance, when resolving *a.b.c* using QMIN resolver (see Fig.1): ❶ The resolver first sends a minimised query containing only the Top-Level Domain (TLD) label *c* to the root server and the root server delegates it to TLD by responding with the TLD's NS record [33]. ❷ Since the TLD nameserver is not yet authoritative for the full domain upon receiving the request for the sub-zone *.b.c*, it delegates the query to the Second-Level Domain (SLD) nameserver that can directly resolve *a.b.c*. ❸ Finally, SLD successfully resolves *a.b.c* and response to resolver.

A sub-zone's NS record can share the same suffix as the original QNAME, such as *ns.c*, resolving the sub-zone *.b.c*. However, it may also refer to a different sub-zone. As illustrated in Fig.1, *ns.m* might also delegate the resolver to the sub-zone *ns.b.c*. We formally define the specific label that leads to a delegation to another nameserver as the **delegation point**. Furthermore, a sub-zone that shares the same suffix as its parent zone is termed **in-bailiwick**, while one with a different suffix is referred to as **out-of-bailiwick**.

2.2 Trade-off between Security and Privacy

While QMIN can enhance user privacy, it increases the resolvers' operational overhead by issuing more iterative queries than traditional resolutions [5, 6]. To mitigate this redundant burden, RFC 9156 [11] mandates specific limits on the iterative process. QMIN implementations must define a maximum iteration count (we denote as *max_iter*), which strictly limits the total number of minimising queries a resolver can perform to fully resolve any single name.

Despite this iteration limit on a single QNAME, attackers can still launch amplification attacks by increasing the number of QNAMEs

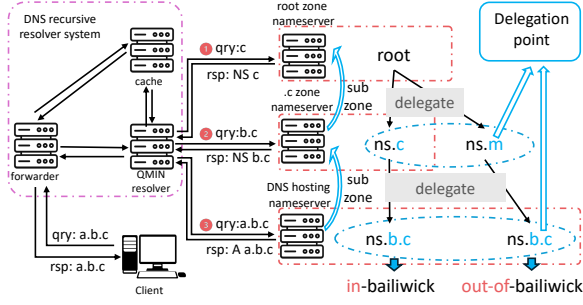


Figure 1: QNAME minimisation recursive resolution.

that each client query needs to QMIN resolve. It has been identified as a potent amplification primitive [17]. This amplification attack can be achieved by exploiting referral responses to generate a significantly larger number of QMIN resolutions than the normal volume. Specifically, an attacker can configure malicious authoritative nameservers to respond with new referrals that deliberately exclude corresponding glue records during DNS resolution. Since these referrals point to nameservers with unknown IP addresses, the resolver is then forced to generate new, separate queries to resolve these referrals. Each of these new resolution tasks introduces a fresh sequence of QMIN-driven queries, effectively compounding the resolving load and enabling a significant traffic amplification against the target. In this paper, we call the QNAME identical to the initial client query as the **original QNAME**, and any other QNAMEs introduced during resolving as **self-introduced QNAMEs**.

To mitigate amplification threat while reducing privacy leakage, two primary optimisation strategies of QMIN have been proposed: 1) restricting QMIN application to the original query name only (*original QNAME only*); and 2) restricting QMIN application to Root and TLD servers only (*root & TLD only*) [21]. However, these strategies reintroduce privacy vulnerability, specifically manifested as unexpected information leakage. As illustrated in our analysis (see Fig. 2), *original QNAME only*, *root & TLD only* may expose sensitive client information when querying for original QNAME or self-introduced QNAME. This is because the full QNAME, rather than the minimised one, is directly sent to resolvers during resolution.

In summary, the trade-off between security mitigation and privacy protection underscores the need for a comprehensive solution. To address this challenge, we first develop a quantification model to assess privacy leakage and the amplification attack potential across all QMIN strategies. We then conduct large-scale measurements to identify the QMIN strategies currently deployed in the wild and evaluate the potential information leakage of popular domain names. Building on our findings, we finally propose a Delegation-Aware QMIN (DAQ) approach to better balance privacy and security in QMIN.

3 Modeling Privacy Leakage and Security Risks

3.1 Modeling Privacy Leakage in DNS Query

We analyze privacy leakage following the Principle of Minimum Disclosure [14]. A query to authoritative server is privacy-preserving if it reveals *only* the necessary subset of the original labels needed for that server's delegation task. QNAME labels sent to a server but

intended for downstream servers are defined as leaked information.

3.1.1 Leakage on a Specific Query. We separately discuss the leakage amount for the original QNAME and the self-introduced QNAME.

Definition of Leakage Amount. We define a single DNS label in the original QNAME as the fundamental unit of information. Consequently, the information of a QNAME is equivalent to its number of labels in the original QNAME or the number of labels from the original QNAME but embedded or represented within labels in a self-introduced QNAME. The information leakage occurs when superfluous labels are exposed to nameservers that do not require them for resolution. For example, the query for *f.g.example.com* shown in Fig. 2 results in 9 units of information leakage during the whole resolving process without QMIN, of which 6 units of information leakage occur when querying the original QNAME (in blue label steps), and 3 units of information leakage when querying the self-introduced QNAME (in red label steps).

Original QNAME Leakage Amount. Suppose a recursive resolver receives a new request querying for domain d which needs to perform iterative queries to n nameservers. The set of these nameservers is denoted as $NS = \{ns_1, ns_2, \dots, ns_n\}$ and the set of private information (*i.e.*, QNAME labels) sent to nameserver ns_i is denoted as s_i . The nameservers in set NS manage different zones and thus the amount of QNAME information they must know also varies. We denote the information that nameserver ns_i must know when resolving as r_i .

So the amount of leaked information in the original QNAME query $O(i)$ shall be the amount of redundant private information in QNAME that exceeds what nameserver ns_i requires, which is denoted as $O(i) = s_i - r_i$. And the total amount of leaked information O during the whole resolving process can be calculated as the sum of leakage in these n nameservers, which can be denoted as $O(d) = \sum_{i=1}^n (s_i - r_i)$. If the resolver sends the full QNAME of the domain d to every nameserver during the resolving process, the private information included in the QNAME is at a maximum amount which can be denoted as S_{max} . And the maximum total leakage can be calculated as $O_{max}(d) = \sum_{i=1}^n (S_{max} - r_i)$. We use the relative privacy leakage amount $O_r(d)$ to measure the degree of unnecessary information leaked by QNAME during the resolution process, which is defined as the ratio of the total privacy leakage to the maximum leakage:

$$O_r(d) = \frac{O(d)}{O_{max}(d)} = \frac{\sum_{i=1}^n (s_i - r_i)}{\sum_{i=1}^n (S_{max} - r_i)} \quad (1)$$

Self-introduced QNAME Leakage Amount. A significant privacy risk emerges when a self-introduced QNAME is delegated to an out-of-bailiwick server, which forces the resolution diverge into another branch of the DNS hierarchy. As shown in Fig. 2, the *example.com* nameserver delegates the query for *g.example.com* to an out-of-bailiwick nameserver *ns.k.g.example.net*, which has 2 units of client privacy (*i.e.*, *g.example*) in risk because the new query starts from the *.net* TLD, which is unaffiliated with the original *.com* zone.

For a self-introduced QNAME with l labels at or after its delegation point, we can denote the privacy amount of each label as a set: $a = \{a_1, a_2, \dots, a_l\}$. Resolver needs to perform iterative queries to extra m nameservers. For each nameserver j , resolver sends l_j

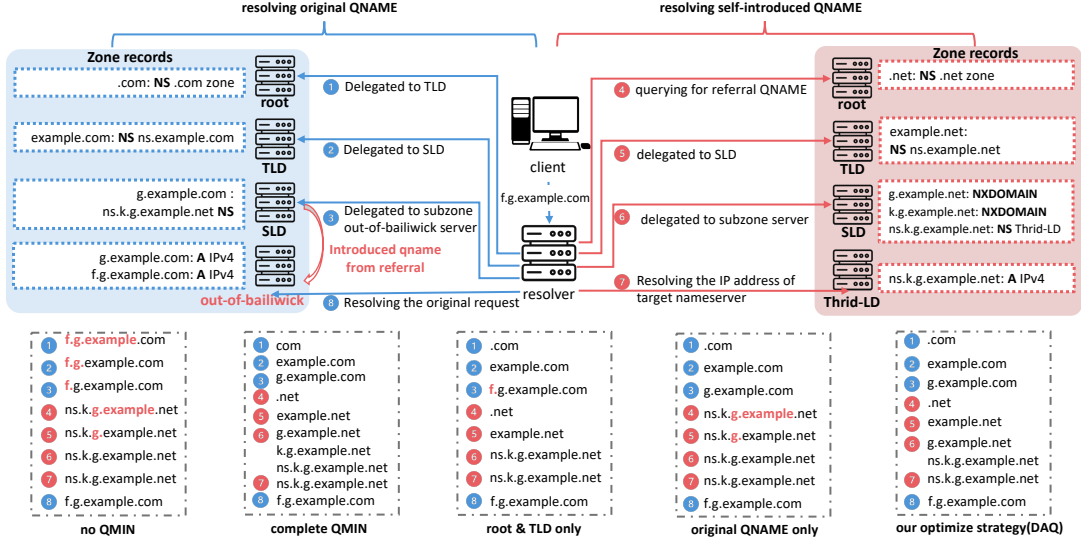


Figure 2: The privacy leakage during resolving QNAME with out-of-bailiwick referral records under different QMIN strategies.

labels of self-introduced QNAME while j only requires a minimum of ln_j labels. So the actual leakage to nameserver j can be calculated as $S(j) = \sum_{k=ln_j+1}^{l_j} a_k$. Then the total amount of leakage during the self-introduced QNAME resolving can be calculated as:

$$S(d) = \sum_{j=1}^m S(j) = \sum_{j=1}^m \sum_{k=ln_j+1}^{l_j} a_k \quad (2)$$

And the maximum leakage amount to nameserver j is the total leakage of all labels that j could potentially resolve: $S_{max}(d) = \sum_{j=1}^m \sum_{k=ln_j+1}^{l_j} a_k$. Finally, we obtain the relative leakage amount:

$$S_r(d) = \frac{S(d)}{S_{max}(d)} = \frac{\sum_{j=1}^m \sum_{k=ln_j+1}^{l_j} a_k}{\sum_{j=1}^m \sum_{k=ln_j+1}^{l_j} a_k} \quad (3)$$

A single DNS resolution can trigger multiple referrals. We next aggregate the leakage across these referrals in different ways by record type (i.e., NS, CNAME, and DNAME), as resolvers handle them differently.

(1) **NS Record Leakage.** When a resolver receives a list of NS records, it typically selects a subset to query. To account for this probabilistic choice, we model the leakage as the average relative leakage amount across all potential NS record resolutions. Given a set of n potential NS records, $NS = \{NS_1, NS_2, \dots, NS_n\}$, the aggregate NS record leakage is:

$$S_{NS} = \frac{\sum_{i=1}^n S_r(NS_i)}{n} \quad (4)$$

(2) **CNAME and DNAME Record Leakage.** CNAME and DNAME records create mandatory referral chains that a resolver must traverse completely. Therefore, we evaluate the cumulative leakage across the entire chain as a whole. For a set of m records in a chain $CD = \{CD_1, CD_2, \dots, CD_m\}$, the aggregate CNAME or DNAME

record leakage is:

$$S_{CNAME} = S_{DNAME} = \frac{\sum_{j=1}^m S(CD_j)}{\sum_{j=1}^m S_{max}(CD_j)} \quad (5)$$

Total Privacy Leakage. Finally, the total information leakage $L(d)$ querying for domain d is the sum of the leakage from the original QNAME and the aggregated leakage from each type of self-introduced QNAME:

$$L(d) = O_r + S_{NS} + S_{CNAME} + S_{DNAME} \quad (6)$$

If a resolver performs *complete QMIN*, $L(d)$ would be 0. However, $L(d)$ will reach a maximum of 4 if the resolver performs *no QMIN* in resolving a domain with an out-of-bailiwick referral chain that includes NS, CNAME, and DNAME records.

3.1.2 Expected Leakage Amount. Given that client queries are inherently probabilistic, we move beyond analyzing the expected leakage amount for an arbitrary query to captures the expected query-level privacy leakage (rather than domain-specific) by overall assessment of top domains.

We model the query probability distribution by leveraging Zipf's law [12, 23], a well-established principle where the popularity of an item is inversely proportional to its rank. Given the strong correlation between a domain's popularity and its query frequency, we use the domain's popularity rank to approximate its query probability, denoted as $P(d)$.

Let the set of domains be denoted as $D(x) = \{d_1, d_2, \dots, d_m\}$. For any given domain $d_j \in D$, its total information leakage $L(d_j)$ can be calculated by using Eq.6 and its request probability is $p(d_j)$. Therefore, the expected information leakage is the sum of each domain's leakage weighted by its corresponding probability and is denoted as $E(L_D)$:

$$E[L_D] = \sum_{j=1}^m p(d_j) \cdot L(d_j) \quad (7)$$

3.2 Modeling Amplification Attack Potential

3.2.1 Attack Workflow. As detailed in Sec 2.2, merely limiting the number of iterative QMIN queries for a single QNAME is insufficient for effective security mitigation. Attackers can instead increase the number of self-introduced QNAMEs that the resolver is forced to process, thereby trapping the resolver in a cascade of intensive QMIN resolution tasks. The core mechanism of this attack hinges on exploiting referral responses in DNS resolution. Specifically, attackers can easily achieve this by continuously generating new referral responses that lack corresponding glue records. When glue records are absent, the recursive resolver must launch excessive resolution processes for these referral domain names before it can resume the original domain resolution path.

As Fig.3, we found that utilizing deep referral recursion is highly effective in maximizing the QMIN reflection amplification potential. Specifically, we control a malicious nameserver to continuously return non-repetitive NS records without glue records, leading the resolver to mistakenly believe that the resolution process is being continuously delegated to different authoritative servers. Upon the recursive delegation depth reaching a predefined depth threshold (e.g., the depth(max) in Fig. 3), the nameserver stops responding with NS records and instead returns a set of random IP addresses. This action compels the resolver to initiate the QMIN process, progressing layer by layer from the QNAME at the final depth back to the original QNAME. Such deep referral recursion with QMIN resolution can significantly increase the resolving load, thereby facilitating substantial traffic amplification against the target.

3.2.2 Theoretical Analysis of Amplification Potential. We begin by formulating the total number of QMIN queries generated in any attack scenario. Let n be the total recursion depth and m_i be the number of parallel query branches at a specific depth i . If each query j at depth i is for a hostname composed of l_{ij} labels, the total number of QMIN queries, N_{total} , is the sum of labels across all queries at all depths:

$$N_{total} = \sum_{i=1}^n \sum_{j=1}^{m_i} (l_{ij}) \quad (8)$$

To model maximum amplification, we consider a worst-case scenario, which is bounded by two key parameters: the maximum recursion depth and the maximum iteration count per QNAME. We denote the maximum iteration count as L_{max} corresponding to the *max_iter* limit referred to in Sec. 2.2. To achieve maximum amplification, the attacker's nameserver is configured to return a new referral for each incoming QMIN query, and thus produce exponential query branches. For instance, at the first level of recursion ($k = 1$), a single query triggers the iterative resolution of a hostname with L_{max} labels. Each of these L_{max} sub-queries then met with a new referral and further creates L_{max} parallel query branches. At the second level ($k = 2$), each of these branches again generates L_{max} sub-queries, resulting in (L_{max}^2) total queries for that level.

Following this iterative process, the maximum number of sub-queries, N_{max} , can be presented as:

$$N_{max} = \sum_{k=1}^n (L_{max})^k \quad (9)$$

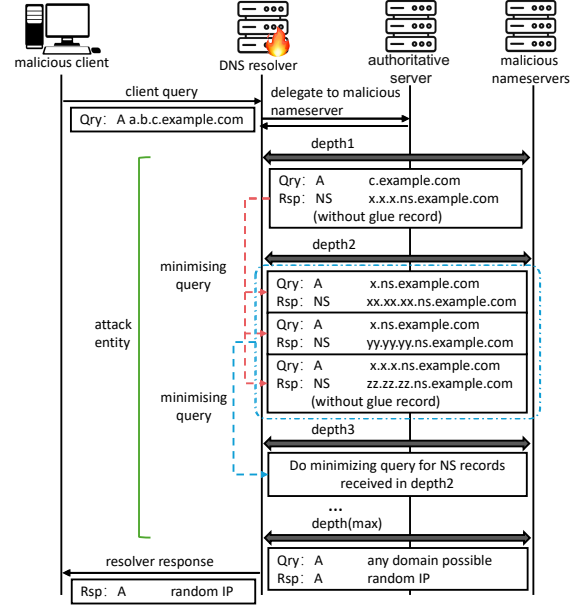


Figure 3: Amplification attack process.

where $L_{max} = 10$ according to RFC9156.

3.2.3 QMIN Effect in Different Strategies. Fig.4 illustrates the theoretical amplification potential across four QMIN strategies: root & TLD only, complete QMIN, original QNAME only, and our proposed DAQ (Sec.5). As *max_iter* increases from 3 to 10 while varying delegation depths (0 to 3), the y-axis is attack effect computed by Eq.(9) which will be larger if the resolver introduces more QNAMEs or does more iterative queries per QNAME. The results consistently demonstrate that complete QMIN suffers from the most severe vulnerability, exhibiting an exponential increase with label depth across all scenarios, peaking at 660 at a delegation depth of 3 (stop increasing because of the output QNAME number limitation). Conversely, the *root & TLD only* strategy shows near-perfect resilience, with its attack effect remaining minimal and largely flat. The *original QNAME only* shows a moderate level of susceptibility, which has a larger effect than DAQ across all tested label numbers and delegation depths.

The attack's potential impact is also severe in the wild, affecting resolvers and nameservers. Indeed, attacks can be launched without self-built nameservers but by exploiting public domains, as complex resolution chains are widespread. We found over 1,800 out of top-1M domains with CNAME chain lengths exceeding 6(max-9). The numerous labels in these chains make the interaction computationally expensive with complete-QMIN: a single client-side query can trigger over 50 authoritative-side interactions by resolver, while this number is 5 times fewer without QMIN. Hence, while self-hosted domains can maximize amplification factor, leveraging multiple such public domains also induces significant traffic pressure.

4 Large-scale Measurement and Analysis

We next move to measure the QMIN deployment and the security and privacy conditions in the wild. Specifically, Sec 4.1 explores the

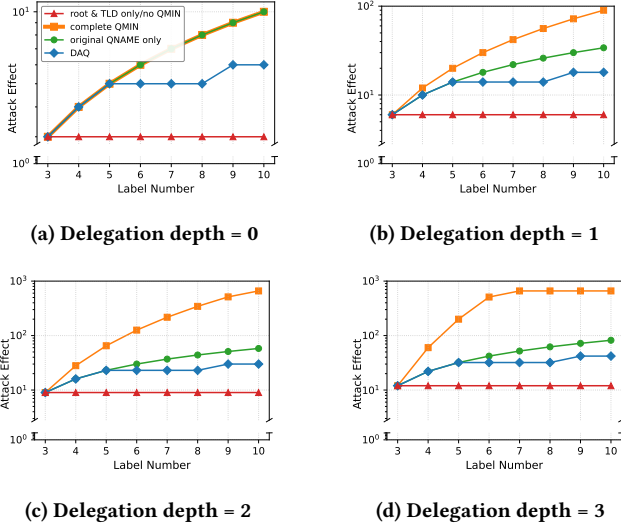


Figure 4: Different QMIN strategies' amplification potential.

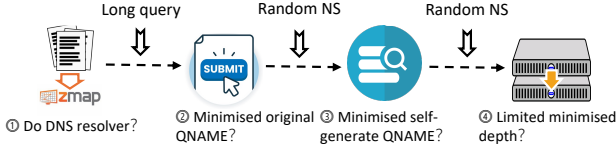


Figure 5: Workflow of the QMIN strategy identification.

QMIN strategy deployment in the wild. Sec 4.2 measures the amplification potential in public resolvers. Sec 4.3 reports the privacy leakage amount among the top 1 million domains. Finally, Sec 4.4 summarizes the balance posture of all these QMIN strategies with respect to privacy and security. Our experimental setup consists of a cloud-based client hosted in Los Angeles, USA, a resolver and an authoritative server deployed on Alibaba Cloud. We implemented the QMIN strategies on our resolver, which is strictly restricted to resolving queries that originate from the local subnet.

4.1 QMIN Deployment in the Wild

The amplification potential is closely related to the QMIN strategies adopted by resolvers; thus, we establish a measurement to characterize resolvers' QMIN strategies as shown in Fig. 5. The measurement begins with ZMap to identify active, open DNS recursive resolvers [27]. For each discovered resolver, we send a query for a domain name that is authoritatively delegated to a nameserver under our control. If we find the resolver performs QMIN on the original QNAME, we then respond with an NS record with a long random suffix but without glue record to test whether QMIN is enabled when resolving referral records (e.g., NS records). If we observe minimisation resolving, we again respond with a referral record with a long random suffix to continue the probe, effectively testing its delegation depth limits. This step-by-step probing allows us to precisely categorize resolvers into four distinct types: (1) do not support QMIN or root & TLD only, (2) perform complete QMIN, (3) original QNAMEs only (and not subsequent NS records), and (4) limit the delegation depth of their minimisation process.

Table 1: The deployment distribution of QMIN strategy.

	complete QMIN	original QNAME	limited depth
#IP	10,828	8,695	2,014
Percentage	50.27%	40.37%	9.36%

We conduct worldwide measurements for open DNS resolvers in the wild to check the support for QMIN on September 20, 2025. Specifically, we get 200,772 active open DNS resolvers by network scanning with Zmap [13, 18]. 21,537 DNS servers (accounting for 10.72%) support QMIN while the other 89.28% servers may only do QMIN to root & TLD servers or are unsupported QMIN at all. As shown in Tab. 1, 50.27% of these resolvers perform complete QMIN, making them potentially vulnerable to an amplification attack. 40.37% of them adopt the strategy of performing QMIN queries only for the original QNAME, while the remaining 9.36% have a limited delegation depth which may result in failing to handle responses that lack glue records.

4.2 Amplification Potential of Public Resolvers

We next focus on amplification attack potential in the wild. While ethical considerations prevented us from measuring the amplification potential of all resolvers, the need to explore real-world amplification potential remains. Thus, we selected 39 popular public resolvers in our measurement. We implement a custom DNS response module to simulate the authoritative-side behavior as the malicious nameservers in Fig. 3 during attack. We completed measurement in April, 2025, and found that QMIN is widely adopted in large-scale public DNS. Our measurement revealed that 30 out of 39 public resolvers adopt QMIN by default (Detailed measuring results are presented in Appendix D). And among the 30 QMIN-deployed resolvers, 21 could have been affected by amplification attack. The remaining resolvers use the *original QNAME only* strategy or the *root & TLD only* strategy. These alternative strategies, by design, avoid being affected by the amplification attack and, consequently, prevent the resolvers from acting as an amplifier in such attacks.

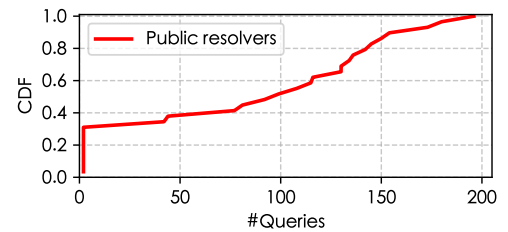


Figure 6: CDF of query numbers for one original query.

Fig. 6 illustrates the CDF distribution of amplification potential of these 30 QMIN-deployed resolvers. The attack leads to a massive increase in the number of self-introduced QNAMEs. For the most-affected resolvers, self-introduced QNAMEs exceed 150, and more than half of these resolvers do queries of more than 100 per request. The Hurricane Electric resolver is the most affected, generating self-generated domains close to 200.

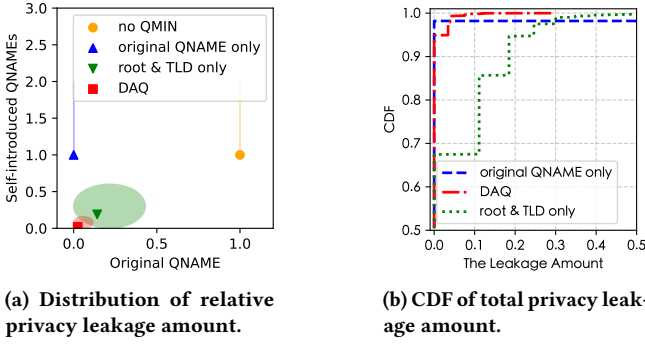


Figure 7: Privacy leakage under different QMIN strategies.

4.3 Privacy Leakage

To conduct a comprehensive assessment of privacy leakage, we designed a large-scale measurement study. This study targeted the top one million domains derived from the Cisco Umbrella dataset [8] and was completed on September 27, 2025.

4.3.1 Self-introduced QNAME Analysis. Tab. 2 presents the quantification of self-introduced QNAMEs and the resulting privacy leakage risk across three key DNS Resource Record (RR) types (NS, CNAME, DNAME) within the Top 1 million domains. The data reveals that CNAME (330,457) is the most prevalent self-introduced QNAME type, closely followed by NS (295,235); DNAME instances are negligible (23). Crucially, we identified 18,024 domains where the referral explicitly contained part of the original QNAME, signifying concrete privacy leakage under restrictive QMIN.

The CNAME mechanism is the dominant contributor, accounting for 14,171 leakage instances and exhibiting a leakage rate of 4.3%. This rate is over three times higher than the NS record rate (1.3%). Although the total DNAME count is low, its leakage rate of 4.3% suggests an equivalently elevated inherent risk profile. These findings emphasize that CNAME referrals pose a disproportionately significant risk factor for unintended domain name exposure when implementing partial QNAME minimisation.

4.3.2 Leakage Amount among TOP 1 Million Domains. Fig. 7a visualizes the privacy leakage distribution across the Top 1 million domains, plotting leakage on the original QNAME (x-axis) against leakage on self-introduced QNAMEs (y-axis) for four QNAME Minimisation (QMIN) strategies calculated by Eq.(1)-(5) introduced in Sec 3.1. The shaded areas represent the distribution range of domain leakage, with the central point indicating concentration.

The complete QMIN strategy (0 for all domains) serves as the theoretical baseline and is omitted. The *no QMIN* strategy represents the upper bound of expected information leakage: since self-introduced QNAMEs are completely unprotected, leakage amounts are distributed between 1.0 and 3.0 (with the average shown closely to 1.0) determined by the number of self-introduced QNAMEs' type number. Conversely, the *original QNAME only* and *root & TLD only* strategies successfully shift their clusters toward (0.0,0.0), confirming their leakage reduction. However, both of them fail to eliminate substantial leakage via self-introduced QNAMEs. The *root & TLD only* strategy, in particular, shows reduced privacy protection when encountering longer domain names.

Table 2: Self-introduced QNAMEs in top-1M domains.

	NS	CNAME	DNAME
Total	295,235	330,457	23
Out-of-bailiwick	3,852	14,171	1

Complementing this domain-level view, Fig. 7b presents the cumulative distribution function (CDF) of the total relative leakage amount across the entire top 1 million domain set using Eq.(6). As previously noted, the *complete QMIN* strategy has a CDF curve simply be a vertical line at a leakage amount of zero and no QMIN curve sits furthest right, confirming maximum cumulative leakage. While the CDF curves for the *original QNAME only* and *root & TLD only* strategies shift remarkably to the left, verifying their overall privacy enhancement, their long non-zero tails confirm the existence of a persistent security gap—a significant fraction of domains continues to incur leakage under these partial approaches.

4.4 Summary

Finally, we summarize the trade-off between security and privacy for four primary QMIN strategies: no QMIN, original QNAME only, root & TLD only, and complete QMIN. Our findings indicate that none of these strategies is effective in achieving a satisfactory balance between the two objectives. Fig. 8 shows that resolving without QMIN and with complete QMIN represent two extremes on the spectrum of privacy versus amplification potential. Resolving without QMIN is resilient to amplification attack where the amplification potential is 12^1 but weakest privacy (expected leakage = 0.051). Conversely, complete QMIN strategy affords the strongest privacy guarantee (expected leakage = 0), but proves highly vulnerable to the attack where the amplification potential is 660. Intermediate strategies present varying trade-offs: *root & TLD only* improves privacy (expected leakage = 0.026) without increasing attack potential, while the *original QNAME only* method further reduces leakage to 0.008 but raises the query count to 84. Therefore, a new strategy is essential to maximizing privacy while ensuring security.

5 Delegation-Aware QMIN

As discussed in Sec.3, privacy leakage arises when upstream nameservers receive clients' query information that they cannot directly resolve and must delegate to another nameserver. If we can precisely identify the delegation points throughout the resolution process, both the privacy leakage and traffic amplification potential will be minimised. Therefore, we seek a solution to predict the delegation points as accurately as possible.

Based on our observations, if a nameserver responds with a record type other than referral (e.g., A type), it suggests that the current nameserver may be authoritative for a larger portion of the QNAME suffix than initially queried. We can also test a larger suffix to identify the true boundary because we are within the delegation point. Conversely, if a referral record is received, it indicates that the currently queried QNAME is beyond the authoritative scope of the nameserver, necessitating delegation to a new server, thus confirming the boundary. Therefore, if we receive a non-referral

¹The number of queries between resolver and authority for a single external request.

Table 3: Delegation point identification strategies overhead.

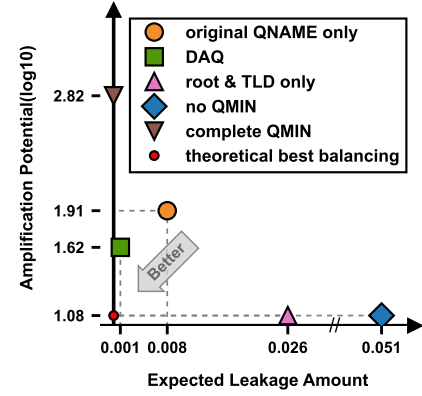
	Complete QMIN	Bisection	Fixed step (size 2)
Time (s)	1.103	1.298	1.394

response, we can send a part of the remaining labels until we either resolve the full QNAME or hit the delegation boundary. However, we cannot simply define this "part" as one label or a full QNAME, as the first one reverts to complete QMIN, leading to a large amplification potential, while the latter results in privacy leakage.

To address this trade-off, we propose a Delegation-Aware QMIN (DAQ) strategy with two choices for identifying the delegation points: the bisection identification and fixed step identification. We present the reason for choosing these strategies in Appendix B. For bisection identification, we iteratively append half of the remaining labels to the QNAME until reaching the delegation point. Conversely, for fixed step identification, we will only append a fixed number of labels in each iteration. Then we integrate this delegation-aware step into the standard QMIN process. Upon receiving a new request, the resolver first attempts to identify the delegation point using one of the candidate strategies. The resolver then constructs and sends a minimised query based on the predicted delegation point. The process proceeds when a response is received: If a referral without glue records is detected, the resolver either introduces a new QNAME (like in complete QMIN) or continues the loop by returning to the delegation-aware step for the new referral. The loop terminates upon receiving the resolution result for the full QNAME, thus completing the resolution process.

We measured the operational efficiency of the bisection and fixed step choices within DAQ. We compared these two methods against the complete QMIN by measuring the time required to resolve the top 1 million domains. We calculated the operation time needed to process these domains and generate minimised QNAMEs by simply counting the processing duration, thus excluding the influence of network environments. As shown in Tab. 3, the bisection identification yields a better result, achieving a resolving time of 1.298 seconds. It is only 0.2 seconds slower than the complete QMIN. We thus employ the bisection identification as the default choice. Alg.1 illustrates DAQ with bisection identification. The detailed description can be found at Appendix C.

We quantified the amplification potential and privacy protection ability of DAQ. As we analyzed in Fig. 4 in Sec.3. DAQ demonstrates a relatively limited amplification potential compared to the complete QMIN and the *original QNAME only* strategies at all delegation depths. And as the recursive depth increases and the maximum iterative label count grows, the benefit of DAQ becomes more pronounced compared to the complete QMIN. The DAQ also demonstrates superior performance in protecting client privacy as shown in Fig. 7. Regarding both the leakage of original QNAMEs and self-introduced QNAMEs, DAQ exhibits the smallest range of privacy leakage, with the average leakage amount close to zero. As to "Root&TLD-only" (same with no-QMIN) and "Original-QNAME-only", while the amplification increases linearly with depth or label number, they sacrifice user privacy. In summary, DAQ exhibits superior performance in balancing security and privacy. Considering the balance between privacy and security of QMIN shown in Fig. 8,

**Figure 8: Security vs. privacy in different QMIN strategies.**

the ideal trade-off is defined by a zero expected leakage amount and an amplification potential of 12. DAQ achieves the minimum distance to this optimal point, with an expected leakage amount of 0.001 and an amplification potential of 42, indicating DAQ has achieved a superior balance between privacy and security.

6 Related Work

QNAME minimisation measurement. Most existing studies focus on measuring the deployment of QNAME minimisation. Vries *et al.* [16] and Magnusson *et al.* [24] separately measured the global deployment of QMIN after RFC7816 [10] and RFC9156 [11]. They found the proportion of open DNS resolvers enabling QMIN had increased year by year, and the use of QMIN worldwide has not significantly burdened the network system with additional traffic. These studies didn't discuss the privacy or security risks of QMIN. **Amplification attack.** The famous NSNXAttack [5] and NRDelegationAttack [6] both exploit NS records to generate a large volume of pseudo-random domain names, inducing a query storm between resolvers and authoritative servers to mount a denial-of-service (DoS) attack. CAMP [17] focuses on a formalized model for DNS and the amplification primitive of DNS. These studies only discussed the security risks QMIN may participate in, leaving the possible privacy leakage issue untouched.

7 Conclusion

QMIN improves users' privacy but introduces amplification threats to resolvers. We develop a quantitative model to assess both the expected information leakage and amplification potential. We conduct large-scale measurements to evaluate the privacy and security conditions for public resolvers and the top 1 million domains. Our findings reveal that none of the existing four major QMIN strategies can effectively strike a balance between privacy and security. To fill this gap, we propose a new QMIN strategy that offers privacy protection close to that of the best method while demonstrating a significantly reduced vulnerability to amplification attacks.

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A Reliability and Ethics.

To avoid causing a negative impact on public resolvers when measuring the amplification of public resolvers, we sent only one request to each of them. This controlled approach caused only hundreds of self-introduced queries in the entire process, a level of activity that has been verified as secure in previous studies [5].

Our methodology of measuring the privacy leakage involves probing for the NS, CNAME and DNAME records associated with each domain. For every domain in our target list, we randomly select a subset of 5 resolvers from the 10,828 QMIN enabled resolvers identified in Sec 4.1 to perform the DNS lookups. In adherence to ethical research guidelines and to avoid imposing an undue burden on the recursive resolvers, we carefully pace our queries. The measurement is executed in batches of 100 domains, with a mandated 1 second sleep interval after each batch is completed.

B Selection of DAQ Step Strategies

To address the trade-off between security and privacy, we propose the Delegation-Aware QMIN (DAQ) strategy. And to select the most suitable approach for DAQ, we initially explored two distinct step adjustment strategies. The first involved an aggressive-to-conservative search where we select a relatively long label subset within the suffix first and then gradually transitions to a more cautious, fine-grained approach. Conversely, the second strategy employed a conservative step wise approach, which maintained a consistently cautious selection of the label subset while attempting to approach the delegation point step-by-step.

Based on these strategies, we selected two candidate methods for comparative evaluation: the Bisection Search and the Fixed-Step Method. The bisection method operates by selecting half of the remaining labels in the current segment and recursively applying this process if the delegation point is not reached.

For the fixed-step method, a step size of 2 was chosen. This decision is supported by empirical data: while the maximum number of labels across the top one million domains is 27, over 860,000 domains have a total label count less than 6. Consequently, a long fixed step is unsuitable. Setting the step size to 2 is twice of complete QMIN approach resulting in a two-fold acceleration.

As shown in Tab. 3, the bisection identification yields a better result, achieving a resolving time of 1.298 seconds. It is only 0.2 seconds slower than the complete QMIN. We thus employ the bisection identification as the default choice.

Table 4: Measurement results for public resolvers.

DNS provider	IP	QMIN support	Attack affected	Queries number
Hurricane Electric	74.82.42.42	✓	✓	196
Safe Surfer	104.155.237.225	✓	✓	180
Dyn	216.146.35.35	✓	✓	173
Yandex	77.88.8.1	✓	✓	154
ControlD	76.76.2.0	✓	✓	150
NextDNS	45.90.28.118	✓	✓	145
SafeDNS	195.46.39.39	✓	✓	142
Norton	199.85.126.10	✓	✓	136
Comodo Secure	8.26.56.26	✓	✓	134
OneDNS	117.50.10.10	✓	✓	134
DNS Forge	176.9.1.117	✓	✓	130
Level3	209.244.0.3	✓	✓	130
Quade	9.9.9.9	✓	✓	116
UltraDNS	156.154.70.1	✓	✓	115
Verisign Public	64.6.64.6	✓	✓	108
SkyDNS	193.58.251.251	✓	✓	99
Freenom Woeld	80.80.80.80	✓	✓	92
AdGuard DNS	94.140.14.14	✓	✓	81
DNS.SB	45.11.45.11	✓	✓	77
OpenDNS	208.67.220.123	✓	✓	44
DNS.WATCH	84.200.69.80	✓	✓	42
114DNS	114.114.114.114	✓	✗	-
CenturyLink	205.171.2.26	✓	✗	-
CleanBrowsing	185.228.168.9	✓	✗	-
CloudFlare	1.1.1.1	✓	✗	-
CZ.NIC ODVR	185.43.135.1	✓	✗	-
LibreDNS	88.198.92.222	✓	✗	-
OpenNIC	103.1.206.179	✓	✗	-
Quad 101	101.101.101.101	✓	✗	-
Strongarm	52.3.100.184	✓	✗	-
360 Secure	101.226.4.6	✗	✗	-
Alibaba DNS	223.5.5.5	✗	✗	-
Baidu	180.76.76.76	✗	✗	-
CIRA Shield	149.112.121.10	✗	✗	-
CNNIC SDNS	1.2.4.8	✗	✗	-
DNS for Family	78.47.64.161	✗	✗	-
DNSPod	119.28.28.28	✗	✗	-
Google	8.8.8.8	✗	✗	-
FreeDNS	37.235.1.174	✗	✗	-

C Delegation-Aware QMIN

Because the bisection identification yields a better result than fixed step, we employ the bisection identification as the default choice. And alg.1 illustrates the detailed process of DAQ with bisection identification. When the resolver receives a new request or is querying for privacy-sensitive labels within self-introduced QNAMES, DAQ performs bisection QMIN if the delegation point of the current nameserver has not been reached or if an attack is suspected (lines 2-5). Otherwise, it sends a sub-query that excludes privacy labels until it reaches the nameserver capable of resolving those labels (lines 6-7), or it directly sends the full QNAME once the bisection QMIN is finished (lines 8-9).

Algorithm 1: Delegation-Aware QMIN(DAQ)

Input : response_type: Type of DNS response; query_level: Current depth; privacy_distribution: Privacy in each label; L_r : labels are resolving; L_p : labels include privacy.

Output: Generated new query.

```

1 Function DAQ:
2   if (query_level == 0)  $\vee$  ( $L_r \in \text{privacy\_distribution}$ ) then
3     if (response_type  $\neq$  REFERRAL)  $\vee$  (attack_suspect) then
4        $L_{\text{waiting}} \leftarrow L_p - L_r$ 
5        $L_r \leftarrow L_r + \lceil L_{\text{waiting}} / 2 \rceil$ 
6     else
7        $L_r \leftarrow L_r + 1$ 
8   else if  $L_r < \text{LOWER\_BOUND}(\text{privacy\_distribution})$  then
9      $L_r \leftarrow \text{LOWER\_BOUND}(\text{privacy\_distribution})$ 
10  else
11     $L_r \leftarrow \text{full\_qname}$ 
12  return
13  GenerateQuery ( $L_r$ )

```

D Public Resolvers

we selected 39 popular public resolvers in our measurement. We completed measurement in April, 2025, and found that QMIN is widely adopted in large-scale public DNS. The results are summarized, with measurement data in Tab. 4. Our measurement revealed that 30 out of 39 public resolvers adopt QMIN by default. And among the 30 QMIN-deployed resolvers, 21 could have been affected by amplification attack. The remaining resolvers use the *original QNAME only* strategy or the *root & TLD only* strategy. These alternative strategies, by design, avoid being affected by the amplification attack and, consequently, prevent the resolvers from acting as an amplifier in such attacks.